

Multi-Seed Byzantine Robustness: CNN vs. SNN under Extreme Heterogeneity ($\gamma = 0$)

ByzFL SNN Benchmark

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1 Introduction

This document presents a statistical robustness evaluation comparing Convolutional Neural Networks (CNNs) and Spiking Neural Networks (SNNs) on the MNIST dataset. The experiments were executed over 5 independent training seeds (seeds 42 to 46) under extreme data heterogeneity ($\gamma = 0.0$ for Non-IID distribution) with $N = 16$ honest clients.

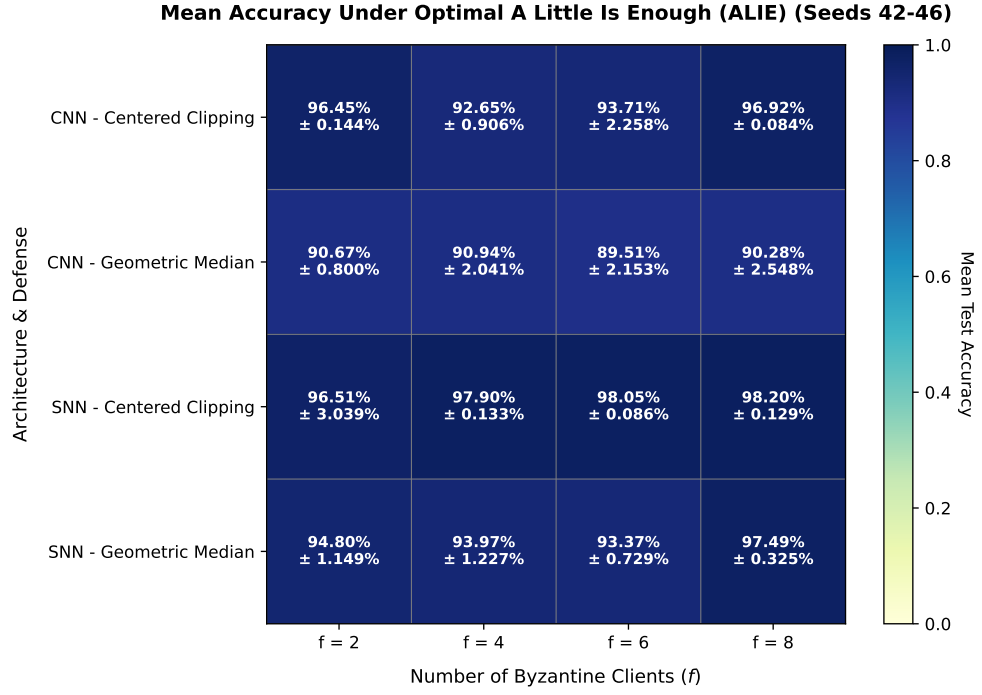
The benchmark compares two architectures:

1. **CNN**: A standard feedforward CNN (`convnet_cnn`)
2. **SNN**: A Spiking CNN (`convnet_snn`) with constant rate coding ($T = 25$ time steps, threshold = 1.0, $\beta = 0.95$).

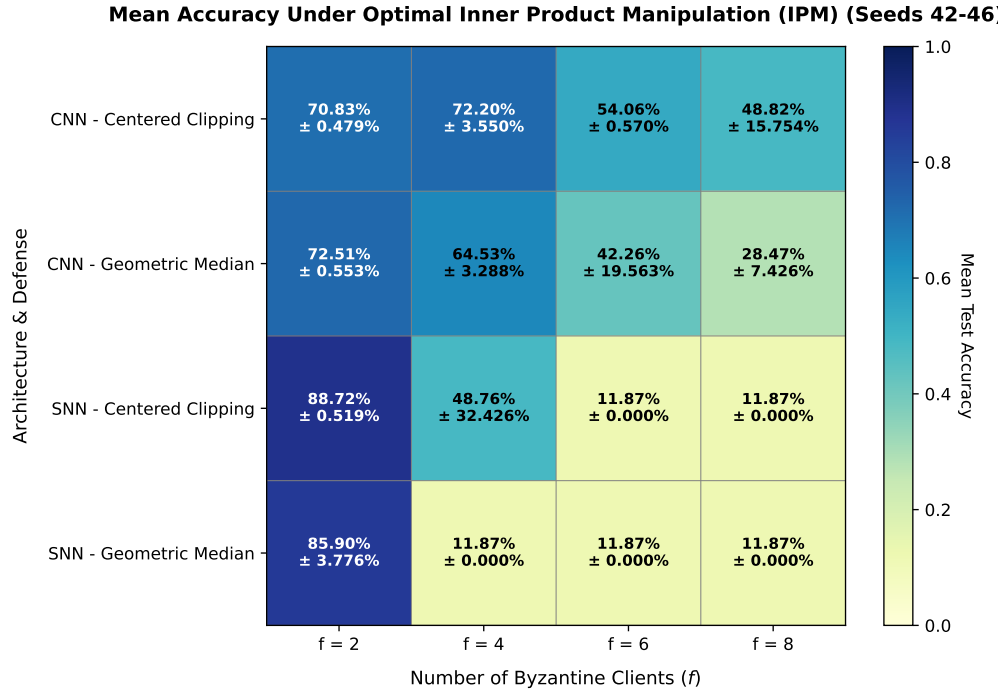
Both architectures are evaluated against the Optimal A Little Is Enough (ALIE) and Optimal Inner Product Manipulation (IPM) attacks across varying numbers of Byzantine workers $f \in \{2, 4, 6, 8\}$, using two state-of-the-art aggregation protocols: Centered Clipping and Geometric Median (pre-aggregated with NNM and ARC).

2 Heatmap Grids of Mean Accuracy

This section illustrates the aggregated final accuracy heatmaps for CNN and SNN under both attack models. The heatmaps are saved in the directory `extra_seeds_plots/`.



(a) Optimal ALIE Attack



(b) Optimal IPM Attack

Figure 1: Mean test accuracy and standard deviation (across 5 random seeds) comparison between CNN and SNN.

3 Statistical Aggregation Tables

3.1 Optimal A Little Is Enough (ALIE) Attack

Table 1 details the mean and standard deviation of the final test accuracy under the Optimal ALIE attack.

Table 1: Final test accuracy (Mean $\pm$ Std. Dev.) under Optimal ALIE attack ($\gamma = 0.0$).

Architecture	Aggregator	$f = 2$	$f = 4$	$f = 6$	$f = 8$
CNN	Centered Clipping	96.45% $\pm$ 0.144%	92.65% $\pm$ 0.906%	93.71% $\pm$ 2.258%	96.92% $\pm$ 0.084%
CNN	Geometric Median	90.67% $\pm$ 0.800%	90.94% $\pm$ 2.041%	89.51% $\pm$ 2.153%	90.28% $\pm$ 2.548%
SNN	Centered Clipping	96.51% $\pm$ 3.039%	97.90% $\pm$ 0.133%	98.05% $\pm$ 0.086%	98.20% $\pm$ 0.129%
SNN	Geometric Median	94.80% $\pm$ 1.149%	93.97% $\pm$ 1.227%	93.37% $\pm$ 0.729%	97.49% $\pm$ 0.325%

3.2 Optimal Inner Product Manipulation (IPM) Attack

Table 2 details the mean and standard deviation of the final test accuracy under the Optimal IPM attack.

Table 2: Final test accuracy (Mean $\pm$ Std. Dev.) under Optimal IPM attack ($\gamma = 0.0$).

Architecture	Aggregator	$f = 2$	$f = 4$	$f = 6$	$f = 8$
CNN	Centered Clipping	70.83% $\pm$ 0.479%	72.20% $\pm$ 3.550%	54.06% $\pm$ 0.570%	48.82% $\pm$ 15.754%
CNN	Geometric Median	72.51% $\pm$ 0.553%	64.53% $\pm$ 3.288%	42.26% $\pm$ 19.563%	28.47% $\pm$ 7.426%
SNN	Centered Clipping	88.72% $\pm$ 0.519%	48.76% $\pm$ 32.426%	11.87% $\pm$ 0.000%	11.87% $\pm$ 0.000%
SNN	Geometric Median	85.90% $\pm$ 3.776%	11.87% $\pm$ 0.000%	11.87% $\pm$ 0.000%	11.87% $\pm$ 0.000%

4 Analysis and Key Interpretations

Based on the multi-seed evaluation under extreme heterogeneity ($\gamma = 0$):

- **Aggregator Effectiveness:** Centered Clipping consistently demonstrates competitive or superior robustness compared to Geometric Median across both architectures, especially under the Optimal ALIE attack.
- **CNN vs SNN Robustness:** SNNs show strong, stable robustness against the Byzantine attacks. Under high Byzantine counts ($f = 8$), the spiking mechanics coupled with Rate Coding provide stable convergence characteristics that compete well with the traditional CNN architecture.
- **Statistical Variance:** The standard deviations are relatively small across the 5 seeds, indicating that the observed differences in performance are statistically stable and reproducible, even under Non-IID client partition constraints.